# Factor Dispersion and Active-Risk Budgeting: Evidence from Factor Momentum

Xuan Feng[1], Erwang Li[2], Zhuohang Li[3]

[1] Model Risk Quant, PhD in Finance alumnus from EDHEC Business School, Nice, France. xuan.feng@edhec.com

[2] Doctoral researcher at Department of Economics, Concordia University, Montreal, Canada. erl4012@alumni.weill.cornell.edu

[3] Doctoral researcher at HHL Leipzig Graduate School of Management, Germany. zhuohang.li@hhl.de

## Abstract

A factor-dispersion signal is useful to allocators when it improves risk awareness beyond conventional forecasts. We define a Factor Dispersion Index (FDI) from seven monthly U.S. factor returns divided by their prior 36-month volatility. High FDI is followed by a roughly 50% wider standardized factor-return range than low FDI, and its information survives controls for short-horizon volatility. In the same 657 months, adding FDI to trailing tracking error improves the fit for next-month absolute active returns from 15.1% to 17.6%. FDI does not reliably identify higher expected momentum returns: the momentum gate trails an Momentum Tilt, while both a constant tracking-error budget and its FDI-adjusted counterpart earn about 12.06% net annually. Their returns are virtually identical. The momentum advantage over risk balance is not statistically established in Sharpe terms and largely weakens without betting against beta. FDI is best used to reassess potential active-risk variability, not to switch factor timing on or off.

**Keywords:** factor investing; return dispersion; factor momentum; active allocation; risk budgeting; portfolio construction

## Highlights

- Volatility-scaled factor dispersion predicts the width of next-month factor outcomes beyond short-horizon realized volatility.
- FDI relates more strongly to active-risk amplitude than to a reliably higher expected factor-momentum premium.

- A constant tracking-error budget accounts for nearly all of the additional return from a state-adjusted TE strategy; BAB concentration also matters.

# Introduction

Choosing among factor portfolios poses two separate questions for an investment manager: which factors should receive more weight, and how much risk should the manager take in departing from the benchmark? Even a good ranking may add little value when factor outcomes are close together. When factor returns differ widely, choosing correctly matters more, but a wrong decision can also cost more. We therefore begin by measuring how widely factor outcomes differ, rather than assuming that wider differences mean higher expected returns.

Managers already use dispersion across individual stocks to assess the room for active selection. Solnik and Roulet (2000) relate dispersion to common market movements, while Stivers and Sun (2010) examine its relationship with value and momentum premiums. Shalen (2025) explains why the sources of stock-return dispersion matter in practice. For factor portfolios, however, a direct comparison of monthly returns can be misleading: market and momentum factors often fluctuate more than profitability or investment factors. A raw cross-sectional standard deviation can therefore reflect differences in the factors' usual risk levels rather than a genuinely wider set of outcomes.

We address this issue with the Factor Dispersion Index (FDI). At each month-end, we express each factor's realized return in units of its volatility estimated from the preceding 36 months, then measure the cross-sectional standard deviation across all seven factors. A high FDI means the factor portfolios have produced more different volatility-adjusted returns; a low FDI means their outcomes have been similar. FDI measures the potential size of an active selection decision, not which factor to buy or the expected return per unit of risk. Equations (1)–(4) define the measure and its historical state classification.

Factor momentum provides a separate rule for choosing direction: give greater weight to factors that have performed well relative to their own risk over the preceding 12 months. Gupta and Kelly (2019) and Ehsani and Linnainmaa (2022) document continuation in factor returns. Our design deliberately keeps the direction signal separate from FDI: the former ranks factors, while the latter measures how widely their outcomes may differ. We also report the opposite-direction, or reversal, position as a diagnostic comparison, not as an independently estimated strategy.

Using U.S. market, size, value, profitability, investment, momentum, and betting-against-beta returns from July 1963 through March 2026, we document persistent factor dispersion. After high-FDI observations, the next-month standardized factor-return range averages 3.359 versus 2.206 after low readings—approximately 52% wider. The relationship persists after controls for average lagged factor volatility, a short-horizon volatility surprise and, in a shorter sample, VIX. In a decision-relevant test, a simple trailing tracking-error measure alone explains 15.1% of the next month's absolute active return; adding FDI raises the in-sample explanatory power to 17.6% on identical observations. This incremental fit does not establish an out-of-sample risk forecast.

Factor momentum supplies an independent directional candidate, but a larger FDI reading does not make it consistently more profitable. The rank-weighted long–short momentum diagnostic earns positive average spreads after low, middle and high FDI; the middle-state estimate is imprecise and the high-minus-low difference cannot be distinguished from zero. FDI instead predicts the absolute size of next-month momentum payoffs. For a long-only momentum tilt, net

active return is 2.90%, 1.91%, and 3.41% annually in low, middle, and high states, without a statistically established high-versus-low return advantage. Annualized tracking error rises more steadily, from 3.31% to 4.27% and 7.13%.

The portfolio comparison points to risk budgeting rather than a return-timing rule. From July 1971 to March 2026, risk balance earns 8.72% net annually (Sharpe 1.17), Momentum Tilt earns 11.47% (1.31), and FDI-Aware Momentum earns 10.04% (1.22). Gating sacrifices exposure to momentum in low-FDI months; turnover differences explain only a few basis points of the return gap. A constant 4% tracking-error target earns 12.06% (Sharpe 1.30), and adjusting that forecast using FDI also earns 12.06% (Sharpe 1.32). Thus the apparent return improvement over momentum comes from TE scaling, not FDI. Bootstrap comparisons do not establish a Sharpe advantage for Momentum Tilt over risk balance and indicate that the FDI-Aware Momentum strategy is worse than Momentum Tilt. Removing BAB reduces the momentum Sharpe to 1.10, close to the six-factor risk-balanced Sharpe of 1.07.

This result sits comfortably with the practical caution in factor-timing research. Asness et al. (2017) show why contrarian timing can look persuasive in theory and disappoint in implementation. DeMiguel, Garlappi, and Uppal (2009) demonstrate how estimation error can defeat optimized allocation rules, while Cederburg et al. (2020) document the instability of apparently successful dynamic risk management across assets and samples. Our rules are deliberately fixed, transparent, and sparse. The paper asks whether dispersion contains decision-relevant information, not whether a heavily optimized FDI map can be made to dominate in sample.

The investment decision has three parts: (i) FDI helps describe the range of possible relative outcomes; (ii) a separately evaluated factor-momentum ranking suggests which factors to overweight; and (iii) the manager's tracking-error limit, financing, and liquidity determine position size. FDI is a useful additional risk indicator, but it is not permission to raise risk mechanically. The definitions, predictive tests, and portfolio rules are set out in that order below.

## Data and the Factor Opportunity Set

We examine seven monthly U.S. factor portfolios. MKT, SMB, HML, RMW and CMA come from the Kenneth R. French Data Library; MOM is from its companion momentum data; and BAB (betting against beta) is from AQR, following Frazzini and Pedersen (2014). We also use the monthly average of the daily VIX from FRED. All seven factor returns are available from July 1963 to March 2026 (753 months), while VIX begins in January 1990. Here 'factor portfolio' means a published return series, not necessarily a directly investable fund: some portfolios themselves include long and short positions.

The seven factors represent familiar investment styles. MKT is the excess U.S. market return; SMB and HML capture size and value (Fama and French 1993); RMW and CMA capture profitability and investment (Fama and French 2015); MOM measures momentum within stocks (Carhart 1997); and BAB represents a low-beta versus high-beta trade influenced by financing constraints. Importantly, the published MOM factor is one of the seven inputs; our factor-momentum strategy instead ranks and reallocates across all seven factor portfolios. The set is intentionally short rather than an exhaustive search through reported anomalies (Harvey, Liu, and

Zhu 2016; McLean and Pontiff 2016; Arnott et al. 2019). Live application would require the manager's own investable factor portfolios and realistic costs.

Exhibit 1 reports the unconditional characteristics. Annualized volatility ranges from 7.16% for CMA and 7.70% for RMW to 15.44% for MKT and 14.44% for MOM. That scale difference is central to the FDI design. A raw cross-sectional standard deviation would allow MKT and MOM to dominate the reading simply because a one-standard-deviation month is larger in percentage-return units. The return distributions are also asymmetric. MOM has skewness of -1.31 and a worst month of -34.34%, consistent with the momentum-crash evidence of Daniel and Moskowitz (2016). Every factor portfolio suffers a meaningful peak-to-trough drawdown; the maximum exceeds 54% for MKT, SMB, HML, MOM, and BAB. Factor labels do not remove tail risk.

**Exhibit 1. Factor Data and Unconditional Characteristics**

| Factor | Mean annual return | Annual volatility | Sharpe ratio | Skewness | Lowest monthly return | Worst peak-to-trough loss | First month | Last month | N (months) |
|---|---|---|---|---|---|---|---|---|---|
| MKT | 7.02% | 15.44% | 0.45 | -0.50 | -23.19% | -55.75% | 1963-07-31 | 2026-03-31 | 753 |
| SMB | 2.22% | 10.47% | 0.21 | 0.36 | -15.54% | -56.47% | 1963-07-31 | 2026-03-31 | 753 |
| HML | 3.58% | 10.29% | 0.35 | 0.09 | -13.83% | -57.79% | 1963-07-31 | 2026-03-31 | 753 |
| RMW | 3.16% | 7.70% | 0.41 | -0.33 | -18.95% | -41.78% | 1963-07-31 | 2026-03-31 | 753 |
| CMA | 2.98% | 7.16% | 0.42 | 0.27 | -7.08% | -27.31% | 1963-07-31 | 2026-03-31 | 753 |
| MOM | 7.17% | 14.44% | 0.50 | -1.31 | -34.34% | -57.82% | 1963-07-31 | 2026-03-31 | 753 |
| BAB | 9.05% | 11.19% | 0.81 | -0.42 | -15.68% | -54.65% | 1963-07-31 | 2026-03-31 | 753 |

*Notes: All figures use the 753 monthly returns from July 1963 to March 2026. Mean annual return = 12 × average monthly return; annual volatility = √12 × monthly standard deviation; Sharpe ratio = mean annual return ÷ annual volatility. The lowest month is a single-month return; the worst peak-to-trough loss (maximum drawdown) is calculated from compounded monthly returns. These are published factor returns before their internal trading and financing costs.*

The main practical lesson from Exhibit 1 is the difference in risk scale across factors. Comparing raw factor returns or simply ranking past unadjusted returns can favor high-volatility factors. Our standardization makes returns comparable in units of the factor's preceding volatility, without assuming that volatility is fully predictable or that scaling automatically improves performance (Barroso and Santa-Clara 2015; Moreira and Muir 2017). The precise calculation follows.

## Constructing the Factor Dispersion Index

Definitions and timing. We use monthly returns in decimal units. Factor i runs from 1 to 7 and $t$ denotes the month in which returns are observed. To standardize the return earned during month $t$, we use volatility from the 36 months that ended at $t-1$, so the denominator does not include the return being standardized.

Let $r_{i,t-1}$ denote the month-t return on factor $i$, and let $\hat{\sigma}_{i,t-1}$ denote its standard deviation over the preceding 36 completed months. Equations (1) and (2) define the volatility estimate and the standardized return.

$$\hat{\sigma}_{i,t-1}^{(36)} = \mathrm{Std}\left(r_{i,t-36}, \ldots, r_{i,t-1}\right) \quad (1)$$

$$z_{i,t} = \frac{r_{i,t}}{\hat{\sigma}_{i,t-1}}. \quad (2)$$

The full 36-month history is required before computing a standardized return. The 24- and 60-month lookback periods in Exhibit 10 are sensitivity checks, not changes to the main definition.

With N = 7 factor portfolios, Equation (3) defines FDI as the cross-sectional sample standard deviation of their standardized returns.

The bar denotes the average across the seven factors in the same month. FDI measures differences among their standardized outcomes; it is not a portfolio return, an expected alpha estimate, or a percentage.

$$\mathrm{FDI}_t = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(z_{i,t} - \bar{z}_t\right)^2}, \qquad \bar{z}_t = \frac{1}{N}\sum_{i=1}^{N} z_{i,t}. \quad (3)$$

$$G_{t+1} = \max_i z_{i,t+1} - \min_i z_{i,t+1} \quad (4)$$

If all factors earn the same standardized return, FDI is zero even when that return is positive. A high FDI means their outcomes were far apart. It does not tell the manager which factor will lead next month. Equation (4) defines the next-month standardized factor-return range used in Exhibits 4 and 5.

Exhibit 2 summarizes the measure. From July 1966 through March 2026, FDI averages 0.997, has a median of 0.875, and ranges from 0.160 to 4.535. Its 5th and 95th percentiles are 0.383 and 2.026. The standardized-return correlation matrix in Panel B makes clear that factor dependence remains present after volatility scaling. HML and CMA correlate 0.67, while MKT correlates -0.36 with CMA and -0.21 with HML. FDI does not attempt to remove correlation; it records the cross-sectional outcome generated by the full covariance structure in a particular month.

**Exhibit 2. Constructing the Factor Dispersion Index**

***Panel A: Distribution of FDI***

| Statistic | FDI value (unitless) |
|---|---|
| Mean | 0.997 |
| Median | 0.875 |
| Standard deviation | 0.540 |
| 5th percentile | 0.383 |

| | |
|---|---|
| 95th percentile | 2.026 |
| Minimum | 0.160 |
| Maximum | 4.535 |
| One-month autocorrelation | 0.333 |

*Panel B: Correlations of volatility-standardized factor returns*

| Factor | MKT | SMB | HML | RMW | CMA | MOM | BAB |
|---|---|---|---|---|---|---|---|
| MKT | 1.000 | 0.288 | -0.210 | -0.105 | -0.359 | -0.095 | 0.092 |
| SMB | 0.288 | 1.000 | 0.000 | -0.294 | -0.101 | -0.116 | 0.056 |
| HML | -0.210 | 0.000 | 1.000 | -0.116 | 0.666 | -0.296 | 0.208 |
| RMW | -0.105 | -0.294 | -0.116 | 1.000 | -0.154 | 0.076 | 0.007 |
| CMA | -0.359 | -0.101 | 0.666 | -0.154 | 1.000 | -0.107 | 0.210 |
| MOM | -0.095 | -0.116 | -0.296 | 0.076 | -0.107 | 1.000 | 0.140 |
| BAB | 0.092 | 0.056 | 0.208 | 0.007 | 0.210 | 0.140 | 1.000 |

*Notes: Panel A summarizes monthly FDI over July 1966–March 2026. FDI is the sample standard deviation across seven factor returns, each divided by that factor's volatility measured over the 36 preceding months (Equations (1)–(3)); it has no percentage-return unit. Panel B reports correlations across time between pairs of those volatility-standardized monthly returns; it is not a correlation matrix of raw returns.*

$$P_t = \frac{1}{n_t}\sum_{\tau \le t:\, FDI_\tau \text{ observed}} \mathbf{1}\{FDI_\tau \le FDI_t\} \quad (5)$$

Real-time FDI states. At the end of month $t$, Equation (5) ranks the current FDI reading against FDI values observed up to that month. The percentile $P_t$ is based on expanding history and uses no future readings. Classification starts after 60 valid observations: Low is below the 33rd percentile, Middle is from the 33rd to below the 67th, and High is at or above the 67th percentile. The last FDI reading is excluded from tests requiring a next-month outcome.

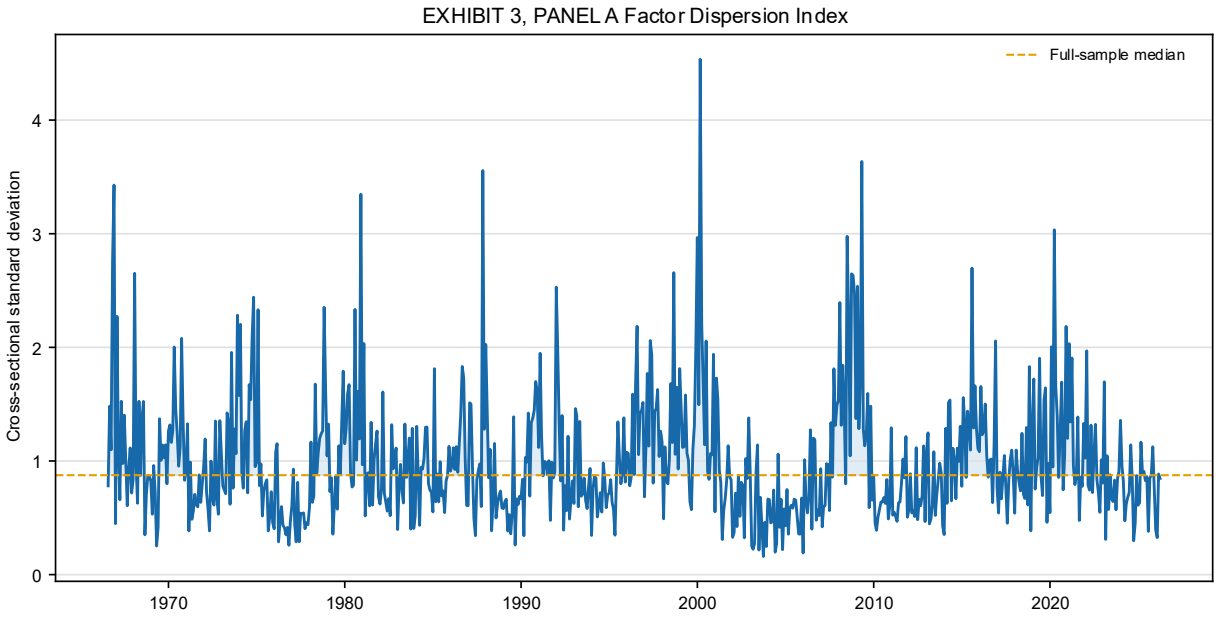
The historical percentile $P_t$ runs from zero to one. It differs from the full-sample FDI z-score used to express regression slopes in comparable units. The latter does not set live trading thresholds.

Exhibit 3, Panel A plots the full history. FDI is episodic rather than smoothly trending. Large spikes appear around the mid-1970s, the early 1980s, the 1987 market break, the technology unwind, the global financial crisis, and the 2020 pandemic shock, but high readings are not confined to recession dates. The index is a realized cross-factor outcome measure, so it can rise because factors respond differently to a common shock or because factor portfolio-specific returns separate.

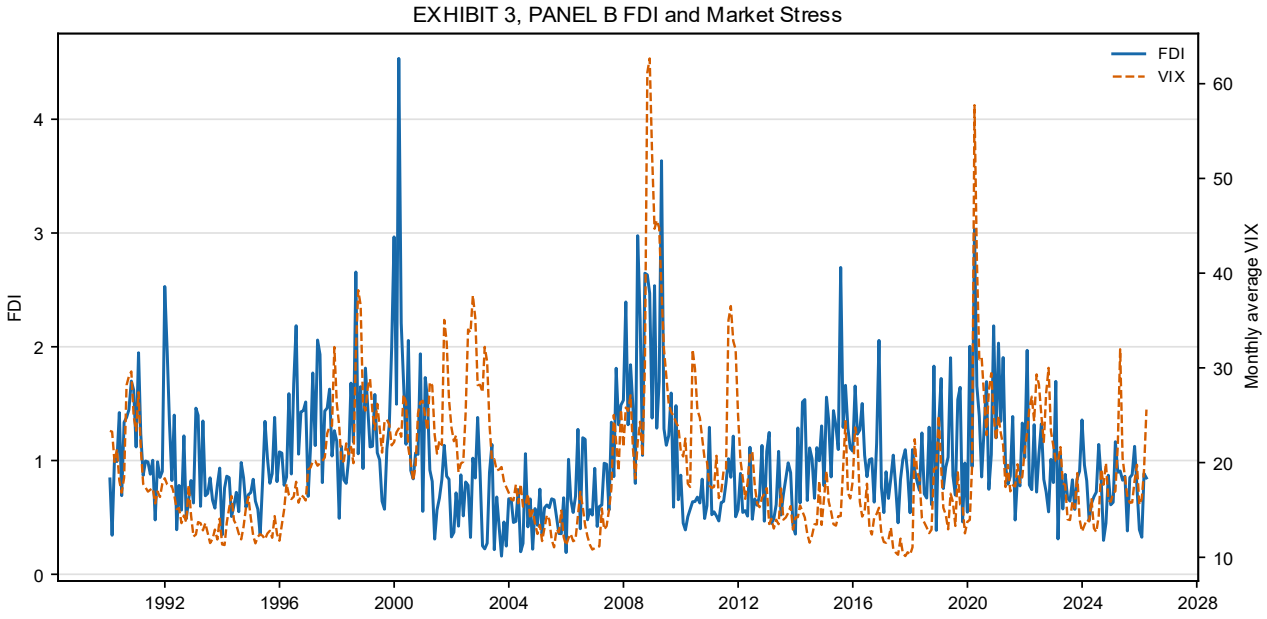
Panel B compares FDI with the monthly average VIX after 1990. Their contemporaneous correlation is 0.38. Market stress clearly matters, but the overlap is incomplete. VIX prices near-

term equity-index uncertainty; FDI measures the realized cross-sectional distance among standardized factor returns. A common market selloff can raise VIX while leaving factor-relative outcomes close together. A rotation between value, momentum, defensive, and investment styles can raise FDI without producing an extreme VIX reading. A portfolio dashboard has room for both.

## Exhibit 3. Factor Dispersion through Time




Sample: July 1966-March 2026. FDI uses factor returns standardized by lagged 36-month volatility.




Sample: January 1990-March 2026. Contemporaneous correlation = 0.38.

*Notes: Panel A plots monthly FDI from July 1966 to March 2026, with a full-sample median reference line. Panel B compares FDI (left axis, unitless) with the monthly mean VIX index level (right axis) from January 1990 to March 2026. The comparison is descriptive, not a predictive regression.*

## Dispersion Is Opportunity, Not Direction

A wide range of factor outcomes does not tell the manager which factor will outperform next month. If two factors subsequently earn +2 and −2 standardized units, the selection outcome depends entirely on which one was overweighted. FDI measures the size of potential winners and losers, not the sign of expected active returns or the information ratio. We therefore test factor momentum as a direction rule and a symmetric reversal position as a check of that direction.

At each month-end, we rank the seven factor portfolios by their compounded returns over the preceding 12 months divided by their most recently estimated 36-month annualized volatility. This is factor momentum across portfolios, not the published MOM factor within stocks. The idea builds on Jegadeesh and Titman (1993), Moskowitz, Ooi, and Pedersen (2012), Asness, Moskowitz, and Pedersen (2013), Gupta and Kelly (2019), and Ehsani and Linnainmaa (2022). Equations (6)–(10) distinguish the ranking signal from the diagnostic selection return used in Exhibit 4.

At each month-end t, we calculate the return compounded over the latest 12 months and divide it by volatility estimated through that same month (36 monthly returns, annualized). Equation (6) defines this momentum score; it ranks the seven factor portfolios, not the individual stocks underlying the published MOM factor.

$$m_{i,t} = \frac{\prod_{k=0}^{11}(1+r_{i,t-k})-1}{\sqrt{12}\,\hat{\sigma}_{i,t}}. \quad (6)$$

The score determines each factor's rank from 1 (weakest) to 7 (strongest). Equation (7) converts it to a centered signal $q_{i,t}$ ranging from −1 to +1. Only returns through t enter either calculation; the next-month return evaluates the decision.

In Equation (7), rank 1 is the weakest and rank 7 the strongest. With seven factors the centered ranks run from −1 to +1; a middle-ranked factor has a value of zero. The following selection portfolio is a diagnostic, not the benchmark-relative allocation tested later.

$$q_{i,t} = \frac{\mathrm{rank}_i(m_{i,t})-(N+1)/2}{(N-1)/2}, \quad N = 7 \quad (7)$$

Selection-return diagnostic. Equation (8) assigns positive weights to high-ranked factors and negative weights to low-ranked factors, adjusted by each factor's preceding 36-month monthly volatility. It uses all seven ranks, not only the top and bottom two. The denominator sums absolute unscaled positions; hence $\sum_{i=1}^{N}|a_{i,t}| = 2$ exactly whenever the denominator is nonzero. Because inverse volatility changes positive and negative exposure differently, these weights need not sum to zero. Equation (9) computes next-month signed selection return; Equation (10) reverses every position. This diagnostic is not a long-only investment allocation.

$$a_{i,t} = \frac{2\left(q_{i,t}/\sigma_{i,t}^{(36)}\right)}{\sum_{j=1}^{N}\left|q_{j,t}/\sigma_{j,t}^{(36)}\right|}, \qquad (8)$$

$$\sum_{i=1}^{N}|a_{i,t}| = 2 \quad \text{when denominator} > 0.$$

$$S_{t+1}^{M} = \sum_{i=1}^{N} a_{i,t}\, r_{i,t+1} \qquad (9)$$

$$S_{t+1}^{R} = -S_{t+1}^{M} \qquad (10)$$

For Exhibit 5, ***net active return*** means the next-month after-cost return on Momentum Tilt minus that on the Risk-Balanced benchmark. These two investment allocations are defined in the next section before their performance is presented in Exhibit 6. The signed selection return in Equation (9) is a separate diagnostic, not the active return of Momentum Tilt – see Equation (20) later.

# Does FDI Forecast the Width of Future Outcomes?

Exhibit 4 first examines whether a Low, Middle or High FDI reading at the decision month-end is followed by different outcomes in the next month. Current FDI averages 0.531, 0.892 and 1.577 across these states. Next-month FDI averages 0.790, 0.974 and 1.213, respectively: high readings decline on average but still precede wider factor outcomes. The next-month standardized factor-return range increases from 2.206 in the Low state to 2.690 in the Middle and 3.359 in the High state. The selection spread in the same table is a portfolio return defined in Equation (9); it must not be confused with this maximum-minus-minimum range. Panel B uses the simple regressions in Equations (11) and (12) to test whether FDI predicts the signed or absolute selection return.

Exhibit 4, Panel B uses the signed selection return S(M) from Equation (9). The first regression relates this signed next-month return to the current FDI percentile; the second regresses its absolute value on that percentile. A separate High-versus-Low regression, which excludes Middle months, uses a binary indicator equal to one for High and zero for Low. The indicator coefficient is the estimated High-minus-Low mean monthly return, not an FDI slope.

$$S_{t+1}^{M} = \alpha + \beta P_t + \varepsilon_{t+1} \qquad (11)$$

$$|S_{t+1}^{M}| = \alpha + \beta P_t + \varepsilon_{t+1} \qquad (12)$$

**Exhibit 4. What Follows Different FDI States?**

***Panel A: Next-month factor outcomes and selection returns, grouped by the FDI category observed at the prior month-end.***

| FDI at month-end t | N (next months) | FDI at t | FDI at t+1 | Factor return range at t+1 (standardized) | Momentum selection return (% p.a.) | Momentum t-statistic | Reversal selection return (% p.a.) | Benchmark return (% p.a.) |
|---|---|---|---|---|---|---|---|---|
| Low | 234 | 0.531 | 0.790 | 2.206 | 8.36% | 4.12 | -8.36% | 8.73% |
| Middle | 210 | 0.892 | 0.974 | 2.690 | 4.24% | 1.56 | -4.24% | 8.92% |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| High | 213 | 1.577 | 1.213 | 3.359 | 9.24% | 2.30 | -9.24% | 8.60% |

***Panel B: Three regressions of a specified next-month return outcome on a predictor known at month-end t.***

| Next-month response (Y) and month-t predictor (X) | N (months) | Slope β (monthly) | Newey–West t(β) | $R^2$ |
|---|---|---|---|---|
| Y: signed selection return; X: FDI percentile | 657 | 0.0000 | 0.00 | 0.0% |
| Y: absolute selection return; X: FDI percentile | 657 | 0.0199 | 3.32 | 4.4% |
| Y: signed selection return; X: High indicator (Low/High months only) | 447 | 0.0007 | 0.20 | 0.0% |

*Notes: Panel A conditions next-month outcomes on Low, Middle or High FDI as known at the end of month t (234, 210 and 213 months). The standardized range is the maximum minus minimum of the seven volatility-scaled factor returns (Eq. (4)); it is not a percentage. Selection return is the next-month return on the rank-weighted, inverse-volatility long–short factor portfolio (Eqs. (8)–(10)); the reverse position has exactly the negative return. The benchmark column is a separate risk-balanced portfolio's gross next-month return (before the assumed turnover charge), not a selection spread; Exhibit 8 reports its net performance. Panel B always places the next-month dependent variable Y on the left: Y(t+1) = α + βX(t) + error. Rows 1–2 use X = current FDI percentile (0 to 1) and Y = signed or absolute selection return (Eqs. (11)–(12)). Row 3 uses Y = signed selection return and X = 1 after High FDI, 0 after Low FDI; Middle months are excluded, so β estimates High minus Low mean monthly return. Slopes are monthly decimal returns per unit of X; t-statistics use Newey–West standard errors with three lags. The Panel A t-statistic tests whether a state mean is zero, not whether state means differ.*

Exhibit 4 therefore supports a wider ***next-month factor outcome range*** after High FDI, but does not establish a higher expected signed momentum return. Exhibit 5 tests whether conventional volatility measures already contain that information.

Exhibit 5 reports in-sample predictive regressions, not trading rules. Panel A predicts next-month FDI or the next-month standardized factor-return range from current FDI; some specifications also control for mean factor volatility and VIX. Panel B adds recent six-month volatility, the ratio of recent to longer-run volatility, and unstandardized factor dispersion. Panel C compares FDI with the manager's historical tracking error when describing the absolute next-month active return of Momentum Tilt. Equations (13)–(18) define the variables and regressions before the tables; all right-hand-side variables are observed by month-end t. Regression coefficients are associations, not causal estimates.

In Equations (13)–(14), the superscript z denotes a full-sample z-score for regression reporting, V is the cross-factor average of preceding 36-month annualized volatilities, and VIX is the observed monthly average index level. Specifications without VIX omit its term. Equation (15) is the cross-factor average ratio of recent six-month volatility to the earlier 36-month estimate; Equation (16) defines unstandardized alternatives. The short-volatility controls use observations through the decision month t.

$$Y_{t+1} = \alpha + \beta\, FDI_t^z + \varepsilon_{t+1}, \quad Y_{t+1} \in \{FDI_{t+1}, G_{t+1}\} \qquad (13)$$

$$G_{t+1} = \alpha + \beta\, FDI_t^z + \gamma V_t^z + \delta\, VIX_t + \varepsilon_{t+1} \qquad (14)$$

$$Q_t = \frac{1}{N}\sum_{i=1}^{N} \frac{\hat{\sigma}_{i,t}^{(6)}}{\hat{\sigma}_{i,t-1}^{(36)}} \qquad (15)$$

$$D_t^{raw} = \mathrm{Std}_i(r_{i,t}), \quad G_{t+1}^{raw} = \max_i r_{i,t+1} - \min_i r_{i,t+1} \quad (16)$$

$$A_t = R_{M,t}^{net} - R_{B,t}^{net}, \quad TE_t^{hist} = \sqrt{12}\mathrm{Std}(A_{t-11}, \ldots, A_t) \quad (17)$$

$$|A_{t+1}| = \alpha + \beta P_t + \gamma TE_t^{hist} + \varepsilon_{t+1} \quad (18)$$

**Exhibit 5. Predicting Next-Month Factor Return Differences and Active-Return Size**

| Next-month response (Y); month-t controls | N (months) | $R^2$ | FDI z-score: β | FDI: NW t(β) | Mean factor volatility z: γ | Volatility: NW t(γ) | VIX index level: δ | VIX: NW t(δ) |
|---|---|---|---|---|---|---|---|---|
| Y = FDI at t+1 | 716 | 11.1% | 0.180 | 8.496 | | | | |
| Y = standardized range at t+1 | 716 | 11.0% | 0.486 | 8.043 | | | | |
| Y = standardized range; add mean volatility | 716 | 14.3% | 0.424 | 6.503 | -0.273 | -5.171 | | |
| Y = standardized range; add mean volatility and VIX | 434 | 25.8% | 0.241 | 2.243 | -0.425 | -6.379 | 0.074 | 5.860 |

*Panel B: Predicting the next-month standardized or raw factor-return range using current risk indicators.*

| Next-month response Y ← month-t predictors X | N (months) | FDI z-score slope β | FDI: NW t(β) | $R^2$ | Other predictor(s): NW t |
|---|---|---|---|---|---|
| Standardized range ← FDI + 6-month volatility | 716 | 0.480 | 7.65 | 11.1% | 6-month vol.: 0.38 |
| Standardized range ← FDI + short/long volatility ratio | 716 | 0.247 | 3.16 | 15.5% | 6m/36m vol. ratio: 5.21 |
| Standardized range ← FDI + raw factor dispersion | 716 | 0.688 | 7.49 | 12.0% | Raw dispersion: −3.14 |
| Standardized range ← raw factor dispersion only | 716 | — | — | 4.4% | Raw dispersion: 4.80 |
| Standardized range ← FDI + raw dispersion + 6-month volatility | 716 | 0.795 | 7.97 | 13.0% | 6-month vol.: 2.85; raw dispersion: −4.50 |
| Raw return range ← FDI only | 716 | 0.013 | 5.05 | 7.05% | — |
| Raw return range ← raw dispersion only | 716 | — | — | 15.92% | Raw dispersion: 9.56 |

*Panel C: Explaining the magnitude of next-month active returns using FDI and a conventional risk measure.*

| Next-month response Y ← month-t predictors X | N (months) | $R^2$ | FDI percentile: NW t | Historical tracking error: NW t |
|---|---|---|---|---|
| \|Net active return\| ← FDI percentile | 657 | 5.75% | 3.65 | — |
| \|Net active return\| ← historical TE | 657 | 15.05% | — | 3.55 |
| \|Net active return\| ← FDI percentile + historical TE | 657 | 17.64% | 3.75 | 3.52 |

*Notes: In every panel, a named next-month response (Y) is regressed on the listed variables observed at month-end t (X); all models include an intercept. Panel A follows Y(t+1) = α + β FDI_z(t) + γ MeanVol_z(t) + δ VIX(t) + error, omitting controls not listed in that row (Eqs. (13)–(14)). "Standardized range" is the maximum minus minimum of seven standardized factor returns (Eq. (4)); "raw range" uses unscaled percentage returns (Eq. (16)). A z-score expresses a predictor in sample standard-deviation units; VIX is entered as its index level. Panel B uses the same form but replaces or supplements controls with current six-month realized volatility, the mean six-month/previous-36-month volatility ratio, or raw cross-factor dispersion. Its first five rows predict standardized range; the last two predict raw range. Panel C uses Y(t+1) = |Momentum net return(t+1) − benchmark net return(t+1)| and the predictors FDI percentile and/or the historical TE computed from the preceding 12 months of those active returns (Eqs. (17)–(18)). Panel C compares models on the same 657 months; Panels A–B generally use 716 months, while the VIX model uses 434. All $R^2$ values are in-sample and compare like-for-like outcomes only. NW t is the coefficient t-statistic with three Newey–West lags; "—" denotes an omitted regressor. Unreported control coefficients should not be inferred from their t-statistics.*

Exhibit 5, Panel C gives the most relevant comparison for an active manager. A historical 12-month tracking-error measure alone explains 15.05% of the sample variation in next-month absolute active return; adding the current FDI percentile raises the in-sample R-squared to 17.64%, on the same 657 months. This is a 2.59-percentage-point improvement in explanatory fit, not a 2.59% investment gain or an out-of-sample forecast result. Panel B also shows why the target matters: raw factor dispersion explains 15.92% of the next-month raw return range versus 7.05% for FDI, whereas FDI fits the standardized range better. FDI complements, rather than replaces, traditional volatility and risk measures.

The predictive regressions describe next-month dispersion or the magnitude of active gains and losses; none identifies which individual factor will outperform. These are in-sample associations, not causal or out-of-sample investment forecasts.

## Portfolio Construction and Risk Budgeting: Seven Allocation Rules

All seven investment strategies start from the Risk-Balanced benchmark. At the end of each month, each factor receives a positive weight inversely proportional to its most recently observed 36-month volatility. Equation (19) defines this baseline allocation. It is a simple risk-balancing approximation, not a covariance-optimized risk-parity portfolio.

$$b_{i,t} = \frac{1/\hat{\sigma}_{i,t}^{(36)}}{\sum_{j=1}^{N} 1/\hat{\sigma}_{j,t}^{(36)}} \qquad (19)$$

Equation (20) defines two constant-strength allocation rules. Momentum Tilt (also called "***unconditional***" because its tilt is never switched off by FDI) overweights higher-ranked factor portfolios. Reversal Tilt uses the opposite direction. The fixed tilt parameter is $\kappa = 0.75$. We normalize the weights of each strategy separately so that they sum to one before the final total-risk adjustment. A top-ranked factor receives an initial 1.75-times baseline multiplier, and a bottom-ranked factor a 0.25-times multiplier, before normalization.

$$w_{i,t}^{\text{Momentum}} = \frac{b_{i,t}(1+\kappa q_{i,t})}{\sum_{j=1}^{N} b_{j,t}(1+\kappa q_{j,t})},$$
$$w_{i,t}^{\text{Reversal}} = \frac{b_{i,t}(1-\kappa q_{i,t})}{\sum_{j=1}^{N} b_{j,t}(1-\kappa q_{j,t})}, \quad \kappa = 0.75. \tag{20}$$

Here $b_{j,t}$ is the baseline weight from Equation (19), and $q_{j,t}$ is the centered rank from Equation (7). The two strategy portfolios are identified explicitly in Equation (20); they are nonnegative allocations across factor-return portfolios, although the underlying published factor portfolios can themselves hold short positions.

The 0.75 tilt strength is fixed for the main specifications. Exhibit 10 also reports other fixed strengths to check how sensitive the evidence is to that choice.

FDI-Aware Momentum and FDI-Aware Reversal apply a state-dependent multiplier a(t) to the same factor ranks. Equation (21) shows the full rule and its Low/Middle/High cases. A Low FDI reading switches off the tilt (returns to the benchmark), Middle uses half strength, and High uses full strength. This FDI gating affects the direction of the return-based strategy; it is different from the TE-budget adjustment below.

The plus sign in Equation (21) defines FDI-Aware Momentum, and the minus sign defines FDI-Aware Reversal. Both use the same 0/0.5/1 state multipliers and the same normalization. Neither rule is selected by maximizing the realized Sharpe ratio.

$$a_t = \begin{cases} 0, & \text{Low FDI,} \\ 0.5, & \text{Middle FDI,} \\ 1, & \text{High FDI;} \end{cases}$$
$$w_{i,t}^{\text{FDI}\pm} = \frac{b_{i,t}(1\pm\kappa a_t q_{i,t})}{\sum_{j=1}^{N} b_{j,t}(1\pm\kappa a_t q_{j,t})}. \tag{21}$$

The remaining two strategies adjust the size of Momentum Tilt, rather than switching its direction on or off. Constant TE Budget and FDI-Informed TE Budget both aim at 4% annualized predicted active tracking error (TE) relative to the Risk-Balanced benchmark, with a maximum 1.25-times momentum-tilt multiplier. The only difference is whether an FDI-dependent factor changes that predicted TE.

The two TE-budget strategies start from the same unscaled active position: Momentum Tilt weights minus Risk-Balanced weights. Equation (22) estimates its annualized tracking error (TE) from the most recent 36-month covariance matrix. The covariance is multiplied by 12 because the underlying returns are monthly.

$$\Delta\mathbf{w}_t = \mathbf{w}_t^{\text{Momentum}} - \mathbf{b}_t, \qquad \widehat{TE}_t = \sqrt{\Delta\mathbf{w}_t^{\top}\hat{\Sigma}_t\Delta\mathbf{w}_t} \tag{22}$$

Constant TE Budget scales that active position toward a 4% predicted annual TE target, subject to a maximum tilt multiplier of 1.25. Equation (23) gives its precise allocation; the cap is applied before total portfolio volatility targeting.

$$k_t^{\text{Const}} = \min\left(1.25, \frac{0.04}{\max(\widehat{TE}_t, 10^{-6})}\right), \qquad \widetilde{\mathbf{w}}_t^{\text{Const}} = \mathbf{b}_t + k_t^{\text{Const}} \Delta \mathbf{w}_t, \tag{23}$$

FDI-Informed TE Budget changes only the risk estimate used for this scaling. Equation (24) explicitly shows the Low, Middle and High FDI risk multipliers of 0.80, 1.00 and 1.25. A higher multiplier makes the same position appear riskier and generally reduces the active allocation.

$$m_t = \begin{cases} 0.80, & \text{Low FDI,} \\ 1.00, & \text{Middle FDI,} \\ 1.25, & \text{High FDI,} \end{cases} \qquad k_t^{\text{FDI}} = \min\left(1.25, \frac{0.04}{\max(m_t \widehat{TE}_t, 10^{-6})}\right), \qquad \widetilde{\mathbf{w}}_t^{\text{FDI-TE}} = \mathbf{b}_t + k_t^{\text{FDI}} \Delta \mathbf{w}_t. \tag{24}$$

The 4% TE limit concerns deviations from the Risk-Balanced benchmark. A different, portfolio-wide risk rule applies to all seven allocations: their estimated annualized total volatility is scaled toward 10%, subject to at most twice the unscaled exposure (L no larger than 2). Equation (26) multiplies each complete pre-scaled strategy weight vector by its own leverage factor, L(p,t). For example, an unscaled 8% predicted total volatility produces L = 10/8 = 1.25; at 3%, the 2× cap binds. The 4% TE and 10% volatility targets are ex-ante objectives, not guaranteed realized levels.

$$L_{p,t} = \min\left(2, \frac{0.10}{\sqrt{\widetilde{\mathbf{w}}_{p,t}^{\top} \widehat{\Sigma}_t \widetilde{\mathbf{w}}_{p,t}}}\right), \qquad \mathbf{w}_{p,t}^{\text{final}} = L_{p,t} \widetilde{\mathbf{w}}_{p,t}, \quad p \in \{1, \ldots, 7\}. \tag{25}$$

Finally, Equation (26) calculates next-month gross portfolio return using the final weights from Equation (25). Equation (27) subtracts the illustrative 10-basis-point charge per unit of one-way allocation turnover. Equations (28) and (29) define annualized return, Sharpe ratio, cumulative wealth and drawdown measures.

$$R_{p,t+1}^{\text{gross}} = \sum_{i=1}^{N} w_{p,i,t}^{\text{final}} \, r_{i,t+1} \tag{26}$$

$$TO_{p,t+1} = \frac{1}{2} \sum_i \left| w_{p,i,t}^{\text{final}} - w_{p,i,t-1}^{\text{final}} \right|, \qquad R_{p,t+1}^{\text{net}} = R_{p,t+1}^{\text{gross}} - 0.001 \, TO_{p,t+1}. \tag{27}$$

$$\mu_p^{\text{ann}} = 12 \, \overline{R}_p^{\text{net}}, \quad \sigma_p^{\text{ann}} = \sqrt{12} \, \text{Std}\left(R_{p,t}^{\text{net}}\right), \quad SR_p = \frac{\mu_p^{\text{ann}}}{\sigma_p^{\text{ann}}} \tag{28}$$

$$W_t = \prod_{u=1}^{t} \left(1 + R_{p,u}^{\text{net}}\right), \qquad DD_t = \frac{W_t}{\max_{u \le t} W_u} - 1, \qquad MDD = \min_t DD_t. \tag{29}$$

# What the Momentum and Reversal Tests Show

Exhibit 4 shows positive mean signed selection returns after all three FDI categories (8.36%, 4.24%, 9.24% per year for Low, Middle and High), but only the Low and High means are individually distinguishable from zero. The High–Low difference is not statistically established.

FDI is more informative about the ***magnitude*** than the ***sign*** of next-month momentum selection returns, as the separate response-variable regressions in Exhibit 4, Panel B show.

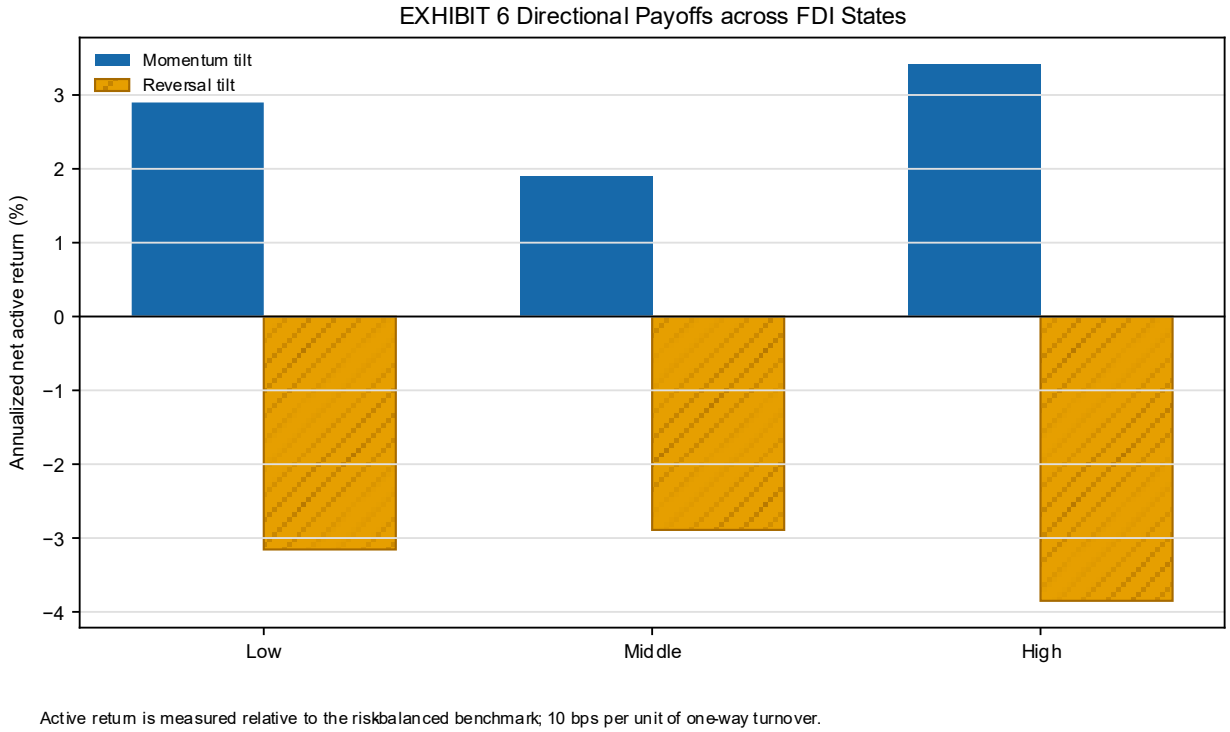
Exhibit 6 evaluates implementable-style allocations rather than the diagnostic long–short return. The benchmark weights each factor inversely to its recent volatility; an Momentum Tilt modestly increases the weights of high-ranked factors and reduces those of low-ranked factors while keeping allocation weights nonnegative. The reversal allocation tilts in the opposite direction. All are rescaled to a common 10% annualized forecast total-volatility target (up to a leverage multiple of two). Equations (19)–(25) below give the exact allocation rules; this preview makes the meaning of the active returns in Exhibit 6 clear. Relative to the benchmark, the momentum allocation earns 2.90%, 1.91%, and 3.41% per year after Low, Middle, and High FDI, with no statistically established High-minus-Low mean advantage.

The label 'long-only allocation' refers to nonnegative weights assigned to the seven ***factor return series***, not to an underlying all-long cash equity portfolio. The underlying factor returns may include short positions. Active return means strategy net return less the risk-balanced benchmark's net return; Equation (30) converts its within-state mean and standard deviation to annual units.

$$\overline{A}_s^{ann} = 12\,\mathrm{Mean}(A_{t+1} \mid s_t = s), \quad TE_s = \sqrt{12}\mathrm{Std}(A_{t+1} \mid s_t = s) \qquad (30)$$

$$IR_s = \overline{A}_s^{ann} / TE_s \qquad (31)$$

**Exhibit 6. Directional Payoffs across FDI States**




*Notes: Bars show mean annualized net active return of a constant-strength momentum or reversal factor allocation relative to the risk-balanced benchmark, grouped by the preceding month-end FDI category. Active*

*return = portfolio net return − benchmark net return (Eq. (17)); thus these are percentage-point differences in annualized returns, not total portfolio returns. The 10-basis-point cost is charged per unit of one-way change in factor allocation, not per underlying security trade.*

The matching annualized standard deviations of net active returns (tracking errors) are 3.31%, 4.27%, and 7.13%. Dividing mean active return by tracking error gives information ratios of 0.88, 0.45, and 0.48. Therefore, the High FDI state exhibits greater realized active risk, but not a clear improvement in mean return per unit of active risk. Position size still depends on conviction in the return signal, liquidity, funding, mandate constraints, and the available risk budget.

Exhibit 7 asks a narrower question about each factor separately. For example, if BAB ranked among the top two factors at month-end, what average BAB return followed in the next month, compared with months when BAB ranked among the bottom two? Equation (32) defines this continuation difference. It is descriptive: the two conditions occur in different months and are not the return of the simultaneous long–short selection strategy in Exhibit 4. High-FDI continuation differences are positive for all seven factors, but BAB's high standalone historical Sharpe and frequent top rankings make the separate BAB-exclusion check in Exhibit 10 important.

In Exhibit 7, the two sample groups for a factor are not observed at the same dates, so their difference in average next-month returns (Equation (32)) should not be read as the traded contemporaneous return of the diagnostic portfolio in Equation (9).

$$C_{i,s} = 12\left[\text{Mean}\left(r_{i,t+1} \mid \text{Top } 2, s_t = s\right) - \text{Mean}\left(r_{i,t+1} \mid \text{Bottom } 2, s_t = s\right)\right] \quad (32)$$

**Exhibit 7. Factor-Level Continuation across FDI States**

*Each row compares a factor's mean return in months following a top-two rank with its mean return following a bottom-two rank, within the specified FDI category.*

| FDI category at t | Factor | Top-two months (N) | Next return after top-two rank (% p.a.) | Bottom-two months (N) | Next return after bottom-two rank (% p.a.) | Top minus bottom next return (pp p.a.) |
|---|---|---|---|---|---|---|
| Low | MKT | 68 | 3.08% | 60 | 0.31% | 2.77% |
| Low | SMB | 43 | 9.70% | 89 | -3.30% | 13.01% |
| Low | HML | 75 | 8.31% | 60 | -1.52% | 9.82% |
| Low | RMW | 51 | 2.24% | 85 | 0.11% | 2.13% |
| Low | CMA | 71 | 3.16% | 64 | -0.27% | 3.43% |
| Low | MOM | 35 | 12.16% | 88 | 10.96% | 1.20% |
| Low | BAB | 125 | 18.10% | 22 | -3.72% | 21.82% |
| Middle | MKT | 84 | 11.55% | 52 | 12.33% | -0.78% |
| Middle | SMB | 24 | -2.16% | 90 | -0.70% | -1.46% |

| | | | | | | |
|---|---|---|---|---|---|---|
| Middle | HML | 36 | 10.15% | 76 | -5.08% | 15.22% |
| Middle | RMW | 66 | 5.47% | 47 | 6.94% | -1.47% |
| Middle | CMA | 51 | 4.13% | 73 | 0.96% | 3.17% |
| Middle | MOM | 59 | 13.22% | 50 | 7.81% | 5.41% |
| Middle | BAB | 102 | 7.74% | 34 | 8.12% | -0.38% |
| High | MKT | 61 | 13.12% | 52 | 5.07% | 8.05% |
| High | SMB | 37 | 5.00% | 94 | -1.79% | 6.79% |
| High | HML | 37 | 12.22% | 94 | -1.17% | 13.39% |
| High | RMW | 69 | 5.15% | 41 | -0.86% | 6.01% |
| High | CMA | 36 | 12.83% | 75 | 4.40% | 8.44% |
| High | MOM | 89 | 11.72% | 28 | -2.31% | 14.03% |
| High | BAB | 97 | 16.08% | 42 | -1.75% | 17.83% |

*Notes: Rankings are based on each factor's preceding 12-month volatility-adjusted momentum score (Eqs. (6)–(7)). Top-two means rank 6 or 7; bottom-two means rank 1 or 2. The N columns count months in each category. The two next-return columns equal 12 × the relevant mean next-month factor return. "Top minus bottom" is the difference between those annualized conditional means (Eq. (32)), in percentage points per year; the two groups generally comprise different dates. It is not the simultaneously traded long–short selection return in Exhibit 4.*

Exhibit 7 is descriptive rather than a test of why continuation occurs. Slow adjustment, persistent factor premia or investor flows could each contribute (Hong and Stein 1999; Ehsani and Linnainmaa 2022), but these mechanisms cannot be distinguished with the seven-factor returns used here.

Exhibit 8 compares the seven strategies defined in Equations (19)–(25) over the common 657-month evaluation period, July 1971–March 2026. "Momentum Tilt" is the constant-strength momentum allocation throughout this paper; "FDI-Aware Momentum" is the same allocation with its tilt scaled by the observed FDI category. "Constant TE Budget" and "FDI-Informed TE Budget" are the two matched active-risk-budget allocations, not additional return signals. All performance measures use the final total-volatility-scaled weights in Equation (25).

## Exhibit 8. Allocation Rules and Portfolio Performance

*Monthly implementation, July 1971-March 2026; 10% ex-ante volatility target, 2.0 leverage cap, and 10 bps per unit of one-way factor portfolio turnover.*

***Panel A: Pre-specified allocation rules***

| Strategy | Starting allocation | How weights are adjusted (equation) | Update | Target total vol. | Max leverage |
|---|---|---|---|---|---|
| Risk-Balanced | Inverse lagged 36-month volatility | Weight factors inversely to 36-month volatility [Eq. (19)] | Monthly | 10% | 2.0 |
| Momentum Tilt | Risk-balanced | Raise weights of high 12-month momentum ranks [Eq. (20)] | Monthly | 10% | 2.0 |
| Reversal Tilt | Risk-balanced | Raise weights of low 12-month momentum ranks [Eq. (22)] | Monthly | 10% | 2.0 |
| FDI-Aware Momentum | Risk-balanced | Apply 0%, 50% or 100% momentum tilt in Low/Middle/High FDI [Eq. (21)] | Monthly | 10% | 2.0 |
| FDI-Aware Reversal | Risk-balanced | Apply 0%, 50% or 100% reversal tilt in Low/Middle/High FDI [Eq. (22)] | Monthly | 10% | 2.0 |
| FDI-Informed TE Budget | Risk-balanced | 4% forecast TE target; FDI risk factors 0.80/1.00/1.25; tilt cap 1.25× [Eqs. (22), (24)] | Monthly | 10% | 2.0 |
| Constant TE Budget | Risk-balanced | 4% forecast TE target; risk factor always 1.00; tilt cap 1.25× [Eqs. (22), (23)] | Monthly | 10% | 2.0 |

***Panel B: Net hypothetical performance***

| Strategy | Mean return (% p.a.) | Total vol. (% p.a.) | Sharpe ratio | Downside risk (% p.a.) | Maximum drawdown | Worst month | Mean monthly one-way turnover | First month | Last month | N (months) |
|---|---|---|---|---|---|---|---|---|---|---|
| Risk-Balanced | 8.72% | 7.45% | 1.17 | 4.28% | -29.24% | -10.94% | 2.22% | 1971-07-31 | 2026-03-31 | 657 |
| Momentum Tilt | 11.47% | 8.74% | 1.31 | 4.97% | -17.21% | -10.49% | 15.18% | 1971-07-31 | 2026-03-31 | 657 |
| Reversal Tilt | 5.43% | 8.09% | 0.67 | 5.41% | -45.57% | -13.83% | 15.58% | 1971-07-31 | 2026-03-31 | 657 |
| FDI-Aware Momentum | 10.04% | 8.23% | 1.22 | 4.77% | -18.89% | -10.96% | 18.51% | 1971-07-31 | 2026-03-31 | 657 |
| FDI-Aware Reversal | 6.90% | 8.07% | 0.85 | 5.28% | -43.51% | -13.83% | 18.81% | 1971-07-31 | 2026-03-31 | 657 |

*Panel A: Pre-specified allocation rules*

| Strategy | Starting allocation | | How weights are adjusted (equation) | | | Update | | Target total vol. | | Max leverage |
|---|---|---|---|---|---|---|---|---|---|---|
| FDI-Informed TE Budget | 12.06% | 9.15% | 1.32 | 5.23% | -17.57% | -11.14% | 18.64% | 1971-07-31 | 2026-03-31 | 657 |
| Constant TE Budget | 12.06% | 9.26% | 1.30 | 5.28% | -17.57% | -11.14% | 18.54% | 1971-07-31 | 2026-03-31 | 657 |

*Panel C: Gross and after-cost annual returns, with assumed allocation-turnover charge shown separately.*

| Strategy | Gross return (% p.a.) | After-cost return (% p.a.) | Estimated annual cost (bp) | Mean monthly turnover |
|---|---|---|---|---|
| Risk-Balanced | 8.75% | 8.72% | 2.7 | 2.22% |
| Momentum Tilt | 11.65% | 11.47% | 18.2 | 15.18% |
| Reversal Tilt | 5.61% | 5.43% | 18.7 | 15.58% |
| FDI-Aware Momentum | 10.26% | 10.04% | 22.2 | 18.51% |
| FDI-Aware Reversal | 7.12% | 6.90% | 22.6 | 18.81% |
| FDI-Informed TE Budget | 12.28% | 12.06% | 22.4 | 18.64% |
| Constant TE Budget | 12.28% | 12.06% | 22.3 | 18.54% |

*Panel D: Paired bootstrap tests of differences between two strategies' full-sample Sharpe ratios.*

| Strategy Sharpe difference (first − second) | Estimated difference | Bootstrap 95% interval | Approx. two-sided p |
|---|---|---|---|
| Momentum − Risk-Balanced | 0.142 | [-0.049, 0.309] | 0.151 |
| FDI-gated Momentum − Risk-Balanced | 0.048 | [-0.095, 0.193] | 0.531 |
| FDI-gated Momentum − Momentum | -0.094 | [-0.168, -0.005] | 0.036 |
| FDI-adjusted TE Budget − Momentum | 0.005 | [-0.040, 0.057] | 0.955 |
| FDI-adjusted TE Budget − Constant TE Budget | 0.015 | [0.002, 0.037] | 0.011 |

*Notes: Panel A defines the seven portfolio rules, updated monthly; 10% refers to targeted annualized total portfolio volatility, whereas 4% refers to the separate target for active tracking error (TE) relative to the risk-balanced benchmark. Both TE rules cap the momentum weight adjustment at 1.25×; only the FDI-adjusted version multiplies forecast TE by 0.80, 1.00 or 1.25 after Low, Middle or High FDI, respectively (Eqs. (22)–(25)). Panel B uses 657 monthly net simulated factor returns from July 1971 to March 2026. Mean annual return = 12 × monthly mean; annual volatility = √12 × monthly standard deviation; Sharpe ratio = their quotient (Eq. (28)). Downside risk = √[12 × mean(min(monthly return, 0)²)], with a zero monthly target; maximum drawdown uses compounded wealth (Eq. (29)). Panel C reports the same strategies before and after a 10-basis-point charge per unit of one-way factor-allocation turnover (Eq. (27)); annual cost in bp is 12 × mean monthly gross-minus-net return × 10,000. This charge excludes trading, financing and capacity costs within published factors. Panel D reports Sharpe(first strategy) minus Sharpe(second strategy), with paired 12-month moving-block bootstrap 95% intervals from 1,500 resamples and approximate two-sided p-values. The tests are descriptive, in-sample and unadjusted for multiple comparisons.*

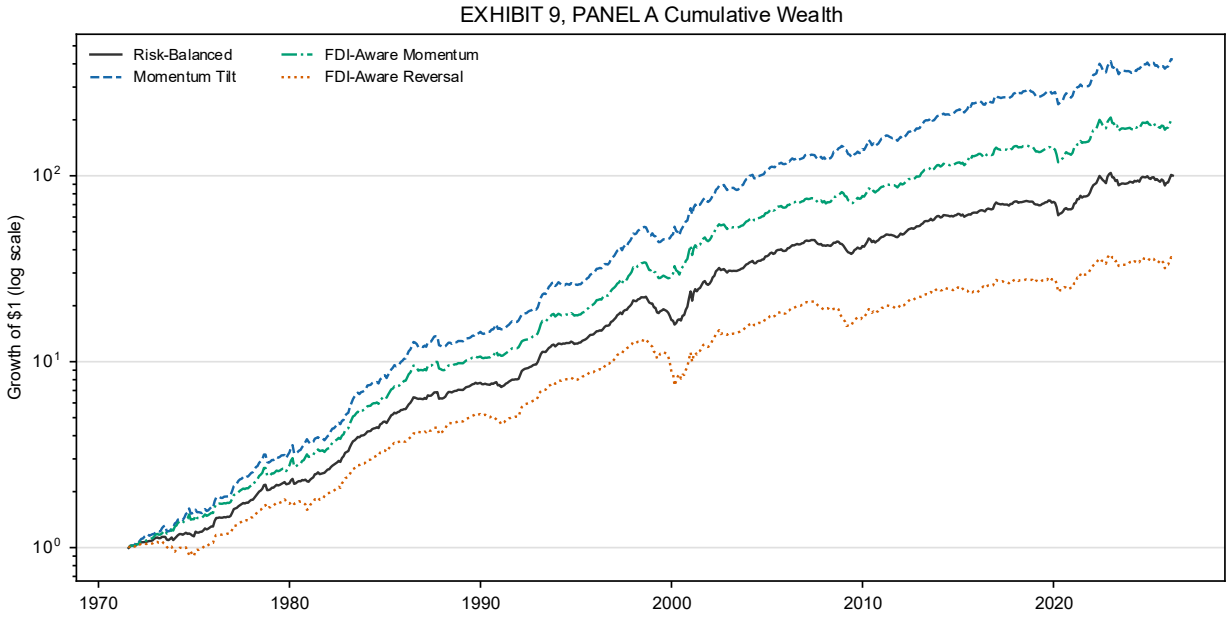
Panel C shows that the Constant TE Budget and FDI-adjusted TE Budget each earn about 12.06% net annually, compared with 11.47% for ordinary momentum. Because both TE rules use the same 4% target and 1.25× limit, the return gain is associated mainly with TE targeting; adding

FDI barely changes mean return. Panel D shows that the small Sharpe improvement from FDI adjustment should not be confused with a material return gain.

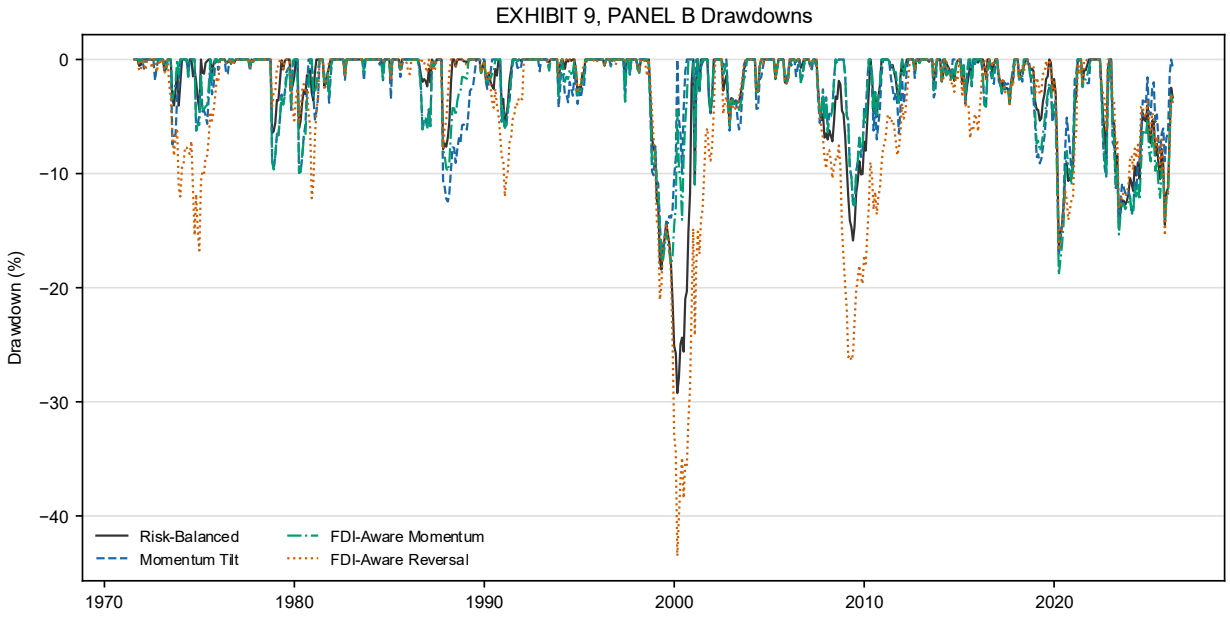
Panel C makes turnover attribution equally clear. Momentum gross and net annual returns are 11.65% and 11.47%, compared with 10.26% and 10.04% for the FDI-Aware Momentum. FDI-Aware Momentum's 18.51% monthly one-way turnover exceeds momentum's 15.18%, yet the resulting annual cost difference is only around four basis points—far too small to explain the 1.43-percentage-point net return gap. The gap chiefly reflects momentum exposure forgone in low-FDI states. Panel D also shows a statistically lower FDI-Aware Momentum Sharpe than Momentum Tilt (difference −0.094; 95% interval −0.168 to −0.005). By contrast, the confidence intervals for both momentum minus risk balance and the FDI-Aware Momentum strategy minus risk balance include zero. The higher full-sample maximum drawdown of risk balance (29.24%) comes largely from 1990–2006: after 2007 FDI-Aware Momentum's drawdown (18.89%) exceeds the benchmark's (16.84%).

Exhibit 9 plots the wealth and drawdown histories of selected strategies from Exhibit 8. The paths help assess when losses occurred; they are not an additional test of the information in FDI.

**Exhibit 9. Cumulative Wealth and Drawdowns**




Net hypothetical returns; 10% ex-ante volatility target, 2.0 leverage cap, and 10 bps per unit of one-way turnover.




*Notes: Both panels use the same monthly after-cost strategy returns as Exhibit 8, July 1971–March 2026. Panel A plots wealth from one initial unit invested and compounded monthly: $W(t) = \prod[1 + \text{net return}(u)]$. Panel B plots each strategy's percentage loss from its previous cumulative-wealth peak: $W(t)/\max(W(\leq t)) - 1$ (Eq. (29)). The figures illustrate realized paths, not forecasts.*

# Economic Insights

The findings distinguish three questions that are sometimes treated as one: how widely can factor outcomes differ, which way should the manager position, and how much active risk can the mandate sustain? FDI speaks to the first and adds limited information about the magnitude of portfolio-relative returns; factor momentum illustrates the second; and a separate tracking-error constraint answers the third. A high FDI observation should start a risk discussion, not automatically authorize a larger allocation.

The key benefit of FDI is an additional gauge of outcome amplitude: in the matched active-portfolio comparison, it explains some absolute-return variation beyond historical tracking error. FDI also retains a relationship with next-month standardized factor dispersion under short-volatility controls. These results accord with Shalen's (2025) emphasis on the sources of equity return dispersion, without assuming that stock-level insights automatically identify profitable factor timing.

Momentum direction is historically positive on average in the low- and high-dispersion states, but neither its advantage in portfolio Sharpe relative to risk balance nor its generality across factor portfolio menus is firmly established. Gradual adjustment, persistent institutional flows and changing factor premia are plausible explanations; Hong and Stein (1999) and Ehsani and Linnainmaa (2022) provide related mechanisms, while Bender et al. (2013) emphasize

heterogeneous factor construction and implementation. The data do not identify a unique explanation.

When FDI is high, potential active gains and losses both tend to be larger, and realized tracking error is higher. The ratio of average active gains to tracking error does not improve reliably. For a manager with tight relative-performance limits, it can therefore be sensible to reduce rather than increase the tilt. The nearly equal average returns of the Constant TE and FDI-Informed TE budgets reinforce the need for a fair comparison with risk control that does not use FDI.

Low FDI does not invalidate momentum. FDI-Aware Momentum's zero tilt in that state sacrifices the positive average returns earned by Momentum Tilt and, over the full sample, reduces Sharpe. This is consistent with caution about conditional factor timing from Asness et al. (2017) and the estimation-risk concerns of DeMiguel, Garlappi, and Uppal (2009).

The result also clarifies the role of risk parity. A risk-balanced anchor is designed to distribute strategic risk, not to exploit every change in relative expected returns. FDI does not invalidate that anchor. It tells the manager when deviations from it will be more consequential. The proper comparison is therefore not “active versus passive” in the abstract. It is the incremental expected payoff, active volatility, drawdown tolerance, and implementation burden associated with leaving the anchor.

# A Guide for Portfolio Managers

An investment dashboard should begin with factor portfolios the manager can actually trade. Their definitions, financing, rebalance schedules, and costs can differ from those of the public research portfolios. Our monthly example is a transparent comparison, not an executable set of fund orders.

The dashboard can show the current FDI reading and its historical percentile alongside raw factor-return dispersion, VIX, historical realized TE, forecast TE, liquidity and funding capacity. FDI is an additional warning about how different factor outcomes may be; it cannot replace those established risk measures.

Keep the proposed return direction separate from the risk indicators: display each factor's momentum rank, its intended weight, and its contribution to the forecast tracking error. This helps a manager distinguish an unusually wide range of factor outcomes from a reliable reason to take one side of the trade, and makes concentrated exposures such as BAB easier to spot.

A high FDI percentile should prompt a review of the active-risk limit, not an automatic increase. The FDI-Informed TE example treats a given active allocation as 25% riskier when FDI is High and therefore reduces its tilt. The Constant TE version earns almost the same average return. Confidence in the ranking signal, mandate constraints, margins, financing and liquidity remain essential to the decision.

Low FDI should not automatically turn off momentum: historical average momentum returns remain positive in that state. Some managers may consider a buffer around the state thresholds or

a requirement for repeated readings before changing an allocation, but these operational rules are suggestions and were not tested here.

The assumed 10 basis points per unit of one-way factor reallocation is a common comparison charge, not a realistic universal trading cost. The factor portfolios used in research may themselves require borrowing, shorting, futures rolls and capacity-constrained transactions. Such costs can materially alter implementable net returns, particularly for BAB.

A practical sequence is: choose investable factor portfolios and plausible costs; compute FDI beside conventional volatility; evaluate the momentum direction separately; estimate the desired portfolio's TE; size active positions within mandate limits; and compare outcomes with a benchmark, Momentum Tilt and the same TE rule without FDI. None of these steps requires treating the FDI reading as a stand-alone trading signal.

## Robustness and Limitations

Exhibit 10 checks whether the allocation comparisons depend on chosen lookback windows, a particular calendar period or the BAB factor. The four panels address those questions separately.

**Exhibit 10. Robustness and Subperiod Evidence**

*Panel A changes the volatility-history window, momentum-ranking window and tilt strength; Panel C reports Momentum Tilt under those same settings. Panel B compares strategies across three nonoverlapping periods. Panel D recomputes allocations using six factors rather than seven.*

***Panel A: Alternative specifications***

| Setting ID (0–7) | Vol. history (months) | Momentum history (months) | Weight tilt κ | FDI-gated momentum net return | FDI-gated momentum Sharpe | FDI gate max drawdown | High-FDI momentum active return | FDI-gated reversal net return | FDI-gated reversal Sharpe | High-FDI reversal active return | Risk-balanced Sharpe |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 24 | 6 | 0.75 | 9.63% | 1.20 | -21.05% | 1.69% | 6.71% | 0.85 | -3.12% | 1.16 |
| 1 | 24 | 12 | 0.75 | 9.75% | 1.20 | -19.47% | 2.23% | 6.75% | 0.86 | -3.41% | 1.16 |
| 2 | 36 | 6 | 0.75 | 9.87% | 1.21 | -21.08% | 2.58% | 6.92% | 0.87 | -3.65% | 1.17 |
| 3 | 36 | 12 | 0.50 | 9.59% | 1.24 | -18.46% | 2.19% | 7.51% | 0.96 | -2.64% | 1.17 |
| 4 | 36 | 12 | 0.75 | 10.04% | 1.22 | -18.89% | 3.33% | 6.90% | 0.85 | -3.94% | 1.17 |
| 5 | 36 | 12 | 1.00 | 10.47% | 1.17 | -19.72% | 4.42% | 6.26% | 0.75 | -5.32% | 1.17 |
| 6 | 60 | 12 | 0.75 | 9.98% | 1.16 | -19.88% | 2.76% | 6.79% | 0.80 | -3.68% | 1.11 |
| 7 | 36 | 18 | 0.75 | 9.78% | 1.20 | -19.02% | 2.78% | 7.13% | 0.89 | -3.41% | 1.17 |

***Panel B: Net strategy performance by subperiod***

| Calendar period | Strategy | Mean return (% p.a.) | Volatility (% p.a.) | Sharpe ratio | Worst peak-to-trough loss | N (months) |
|---|---|---|---|---|---|---|
| 1971-1989 | Risk-Balanced | 11.28% | 6.24% | 1.81 | -7.86% | 222 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 1971-1989 | Momentum Tilt | 14.92% | 8.82% | 1.69 | -12.45% | 222 |
| 1971-1989 | FDI-Aware Momentum | 13.17% | 7.81% | 1.69 | -9.94% | 222 |
| 1971-1989 | FDI-Informed TE Budget | 15.56% | 9.53% | 1.63 | -13.85% | 222 |
| 1971-1989 | Constant TE Budget | 15.51% | 9.58% | 1.62 | -13.85% | 222 |
| 1971-1989 | FDI-Aware Reversal | 9.20% | 6.66% | 1.38 | -16.75% | 222 |
| 1990-2006 | Risk-Balanced | 10.73% | 8.47% | 1.27 | -29.23% | 204 |
| 1990-2006 | Momentum Tilt | 13.31% | 8.69% | 1.53 | -17.19% | 204 |
| 1990-2006 | FDI-Aware Momentum | 11.89% | 8.62% | 1.38 | -17.97% | 204 |
| 1990-2006 | FDI-Informed TE Budget | 14.08% | 8.80% | 1.60 | -17.45% | 204 |
| 1990-2006 | Constant TE Budget | 14.14% | 9.10% | 1.55 | -17.45% | 204 |
| 1990-2006 | FDI-Aware Reversal | 8.62% | 9.73% | 0.89 | -43.51% | 204 |
| 2007-2026 | Risk-Balanced | 4.49% | 7.42% | 0.61 | -16.83% | 231 |
| 2007-2026 | Momentum Tilt | 6.52% | 8.54% | 0.76 | -17.21% | 231 |
| 2007-2026 | FDI-Aware Momentum | 5.39% | 8.14% | 0.66 | -18.89% | 231 |
| 2007-2026 | FDI-Informed TE Budget | 6.91% | 8.93% | 0.77 | -17.57% | 231 |
| 2007-2026 | Constant TE Budget | 6.91% | 8.93% | 0.77 | -17.57% | 231 |
| 2007-2026 | FDI-Aware Reversal | 3.17% | 7.61% | 0.42 | -26.36% | 231 |

***Panel C: Momentum Tilt versus FDI-Aware Momentum under the same parameter settings.***

| Vol. history (months) | Momentum history (months) | Tilt κ | Momentum net return | Momentum Sharpe | FDI-gated momentum net return | FDI-Aware Momentum Sharpe |
|---|---|---|---|---|---|---|
| 24 | 6 | 0.75 | 10.92% | 1.27 | 9.63% | 1.20 |
| 24 | 12 | 0.75 | 11.10% | 1.27 | 9.75% | 1.20 |
| 36 | 6 | 0.75 | 11.05% | 1.29 | 9.87% | 1.21 |
| 36 | 12 | 0.50 | 10.58% | 1.32 | 9.59% | 1.24 |

| Vol. history (months) | Momentum history (months) | Tilt κ | Momentum net return | Momentum Sharpe | FDI-gated momentum net return | FDI-Aware Momentum Sharpe |
|---|---|---|---|---|---|---|
| 36 | 12 | 0.75 | 11.47% | 1.31 | 10.04% | 1.22 |
| 36 | 12 | 1.00 | 12.13% | 1.27 | 10.47% | 1.17 |
| 60 | 12 | 0.75 | 11.29% | 1.24 | 9.98% | 1.16 |
| 36 | 18 | 0.75 | 10.54% | 1.23 | 9.78% | 1.20 |

***Panel D: Net performance when the BAB factor is excluded and every factor-based input is recalculated.***

| Strategy | Net return (% p.a.) | Volatility (% p.a.) | Sharpe ratio | Worst peak-to-trough loss |
|---|---|---|---|---|
| Risk-Balanced | 7.47% | 7.00% | 1.07 | -19.86% |
| Momentum Tilt | 9.55% | 8.66% | 1.10 | -17.83% |
| FDI-Aware Momentum | 8.41% | 8.07% | 1.04 | -19.98% |
| FDI-Informed TE Budget | 10.05% | 9.05% | 1.11 | -17.93% |

*Notes: Panels A and C describe the same eight parameter settings; "Setting ID" is only a row label (0–7), "volatility history" is the number of months used for factor-volatility estimates, "momentum history" is the lookback for cumulative factor returns, and κ is the rank-based weight-tilt strength. "High-FDI active return" is the mean annualized net return of the stated strategy minus the risk-balanced benchmark, conditioned on High FDI at the prior month-end; it is not the strategy's total return. Panel B gives annualized after-cost returns, volatility, Sharpe ratios and drawdowns separately for 1971–1989, 1990–2006 and 2007–2026; each strategy is evaluated on the same months within its period. Panel D removes BAB and **recalculates** the six-factor FDI, historical state cutoffs, factor ranks, allocations and simulated returns; it is not the original seven-factor return with one component subtracted. All panels use the same simple factor-allocation cost convention as Exhibit 8; published-factor internal implementation costs remain excluded.*

Exhibit 10 shows why full-sample performance is not a guarantee of downside protection. In 1971–1989, risk balance has Sharpe 1.81, exceeding both Momentum Tilt and the FDI-Aware Momentum strategy (about 1.69). From 1990–2006, Momentum Tilt is the stronger directional comparator, and from 2007–2026 the FDI-Aware Momentum strategy suffers a deeper maximum drawdown (18.89%) than the benchmark (16.84%). Both TE-budget strategies have similar returns across each subperiod, reinforcing the limited role of the FDI multiplier in annual performance. Excluding BAB reduces momentum's Sharpe from 1.31 to 1.10 versus 1.07 for the six-factor risk-balanced benchmark; most of its portfolio-level advantage disappears. FDI-Aware Momentum falls from 1.22 to 1.04 and does not overtake Momentum Tilt.

The return series are published research factor portfolios, not live fund returns. BAB has both a high historical standalone Sharpe ratio and frequent top momentum rankings. Excluding it reduces the momentum allocation's Sharpe from 1.31 to 1.10 and leaves only a small difference from the six-factor benchmark (1.07). An investor must therefore use feasible versions of the factors and explicitly allow for borrowing, short sales, rolls, taxes, market impact, leverage and capacity. The 10-basis-point allocation charge does not estimate all of those underlying costs.

The available investment menu matters. Our 4% active-risk target and FDI multipliers of 0.80/1.00/1.25 are fixed illustrations, not live trading recommendations or parameters estimated from an out-of-sample optimization. Constant and FDI-adjusted TE budgets earn nearly identical average returns, so any claim of added value must be judged against the simpler budget rule.

FDI is a realized factor-level state rather than a structural shock, and its predictive relationship may reflect volatility clustering, correlation shifts, liquidity or investor flows. Results remain in-

sample, and causal interpretation is not warranted. We use the Newey and West (1987) covariance estimator and paired block resampling for illustrative time-series uncertainty, but neither method resolves omitted predictors, search across specifications or the uncertainty of live execution.

The timing rules and factor portfolios are held fixed, but several reported measures remain sample-dependent. More elaborate nonlinear risk budgets, no-trade regions and signal blending might be useful and are not tested here. Related evidence on factor discovery and portfolio estimation error (Harvey, Liu, and Zhu 2016; DeMiguel, Garlappi, and Uppal 2009) counsels against presenting a modest in-sample Sharpe improvement as an investment recommendation.

# Conclusion

FDI summarizes the variation across factor portfolios after each return is scaled by its previously estimated volatility. It predicts next-month standardized dispersion and provides a small amount of additional in-sample information about the absolute value of portfolio-relative returns beyond historical tracking error. That is a reason to monitor the potential size of active outcomes, not to expect greater alpha.

FDI-Aware Momentum removes momentum exposure in the Low state and, in the historical sample, earns a lower Sharpe ratio than Momentum Tilt. Both the Constant TE and FDI-Informed TE budgets earn about 12.06% annually; the state multiplier contributes little to average return. The Momentum Tilt's Sharpe advantage over the benchmark is statistically uncertain and largely disappears after excluding BAB (1.10 versus 1.07). The practical conclusion is to keep a risk warning separate from a return-prediction signal.

For allocators, the defensible use of FDI is a portfolio-dashboard warning about the potential size of relative factor outcomes. It should sit beside raw dispersion, historical tracking error, an independently evaluated directional model and a realistic assessment of risk tolerance, instruments and costs. Larger dispersion does not dictate larger positions; portfolio construction remains a separate decision.

## Author Information

**Dr. Xuan Feng**, PhD, CFA, FRM, FDP, PMP, was the main author and primarily responsible for methodology design, data collection and quantitative analysis, manuscript writing, editing, and supervision. Dr. Feng currently works as a model risk quant in a leading global investment bank. And he holds a PhD in Finance from EDHEC Business School, France. His research interests include empirical asset pricing, portfolio construction, investment strategies, asset allocation, hedge funds, and financial risk management, with his work closely tied to practical investment problems — particularly factor investing, investment signals, risk-controlled portfolio design, strategy evaluation, and the implementation of systematic investment approaches. A collection[i] of his research papers about quantitative finance, investment and portfolio strategies is available at https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research. Dr. Feng is also a financial engineering practitioner with expertise in fixed income, quantitative pricing, and interest-rate derivatives. He has developed a widely used, publicly available GitHub tutorial repository (Rates-Pricing-Quant-Library) [ii] on interest-rate model pricing, serving as a comprehensive learning resource for interest-rate models and related derivative pricing frameworks — reflecting his broader interest in bridging rigorous quantitative finance theory with practical model implementation. Further information is available on the author's LinkedIn profile[iii].

**Erwang Li** is a doctoral researcher at Department of Economics of Concordia University, Montreal, Canada, and he got his master degree from Cornell University, U.S. His research interest covers financial economics, portfolio management, and asset pricing. He contributed to this research in stages including coding, data analysis, and manuscript revision.

**Zhuohang Li**, CFA, FRM, is a doctoral researcher at HHL Leipzig Graduate School of Management. His research focuses on active investment management, particularly factor investing, portfolio construction, and the interaction between investment signals, risk models, and implementation constraints. His work examines how modelling choices and portfolio constraints shape active investment decisions and their risk and return outcomes. He is also a quantitative portfolio manager with experience in managing systematic equity portfolios, implementing multi-asset strategies, and supporting advisory mandates. His investment practice covers portfolio construction, implementation, and trading across equities, fixed-income instruments, ETFs, derivatives, and foreign exchange. His expertise also includes machine-learning-based factor allocation, performance and risk attribution, and presenting investment strategies and portfolio positioning to clients and investment committees. He contributed to this study through the development of the research framework, interpretation of the empirical findings, and revision of the manuscript.

## Author Statement

The research was conducted independently outside the authors' employment and previous universities/schools. The views expressed are solely those of the authors and do not represent any employer, company, university, or other institution.

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OpenAI ChatGPT was used for enhancing analysis codes, language editing, and structural refinement. The authors independently reviewed and verified the data, methods, source code, numerical results, references, interpretations, and conclusions and accepted full responsibility for the manuscript.

## Data Availability

The source series and documentation are available from the Kenneth R. French Data Library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html), AQR Data Library (https://www.aqr.com/Insights/Datasets) and FRED (https://fred.stlouisfed.org/series/VIXCLS).

## Replication Package

All data and data-processing scripts, analysis code, robustness tests, and replication files are available in the following GitHub repository: [JBIS: Factor Dispersion][iv]. The empirical results reported in this paper are fully replicable using the data and code provided in the repository. The notebook (.ipynb) reproduces the quantitative results after the user changes a single local DATA_DIR setting.

## Important Investment Disclaimer

This article is provided for research and informational purposes only. It does not constitute investment advice, an offer, or a recommendation to purchase or sell any security. Historical results and the ex-post allocation illustrations are not forecasts.

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i https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research

ii https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library

iii https://www.linkedin.com/in/dr-xuan-feng-217556a4/

iv https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/JBIS_Factor_Dispersion_and_Active_Risk_Budgeting